

Safe Reinforcement Learning-Based Transient Stability Control for Islanded Microgrids With Topology Reconfiguration

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Abstract—This paper proposes a safe reinforcement learning (RL)-based transient stability emergency control (TSEC) method for islanded microgrids. RL requires extensive interaction with the environment to learn control strategies, hence, a data-driven approach is used as a substitute for time-consuming time-domain simulation calculations. Deep sigma point processes (DSPP), which is a Gaussian process model, is utilized to predict the normal distribution of transient stability of microgrids and to construct a transient stability chance constraint. Reward-constrained policy optimization (RCPO) can simultaneously achieve objective prediction, policy learning, and constraint cost coefficient update across multiple timescales. RCPO interacts with the DSPP-based microgrid environment through a multi-process parallel manner, greatly increasing the training speed. Case studies on a real islanded microgrid demonstrate that the proposed method can efficiently and quickly obtain the optimal emergency control strategy while adhering to all hard constraints.

Index Terms—Safe reinforcement learning, chance constraint, stability control, PVs, topology change, microgrid.

I. INTRODUCTION

ISLANDED microgrids operate independently from the main grid and are powered by local distributed generation sources, such as diesel generators (DGs), natural gas generators (NGs), PVs, and battery energy storage systems (BESSs) [1], [2]. Due to the lack of support from the main grid in terms of inertia, damping, and spinning reserve capacity, islanded microgrids, when facing large disturbances, such as three-phase short circuit faults and generator outages, may experience transient instability much more rapidly compared to those connected to main grid [3], [4]. Therefore, it is

crucial to efficiently and quickly implement transient stability emergency control (TSEC) after a disturbance occurs.

The most accurate transient stability estimation (TSE) is based on time-domain simulation (TDS) that solves a set of differential-algebraic equations (DAEs), but this is very time-consuming [5], [6]. In addition, TDS is limited by the accuracy of the model. Based on TDS, the trajectory sensitivity method has been applied to emergency control. However, it requires real-time sensitivity calculations, making it time-consuming, and its reliance on linearization introduces control errors [7]. With the widespread deployment of advanced sensors in microgrids, data-driven deep learning methods are widely used, such as deep belief network (DBN)-based islanding detection [8], neural Lyapunov approach-based transient stability assessment [9]. In [10], an evolutionary algorithm (EA) is used to compute the TSEC strategy, which is then learned directly through deep learning to enable rapid policy generation. However, this approach carries the risk of the policy deviating from the safe region. Reference [11] proposes a neural network-based secondary voltage control method for microgrids but its reliance on a small-signal model limits its ability to handle rotor angle stability, which involves stronger nonlinearities. Overall, these methods are mostly based on deterministic estimations and fail to account for the uncertainties in renewable energy sources, loads, and models. Probabilistic estimations based on deep learning can be categorized into two types: one relies on Monte Carlo simulations (MCS) [12], and the other is probabilistic neural networks (PNN), capable of directly outputting the probability distribution of estimations [13]. MCS-based probabilistic estimation requires execution in each control iteration, making it time-consuming. Typical algorithms of PNN include Bayesian neural networks [14] and Gaussian processes (GP) [13]. These can be effectively integrated with various control methods, serving as transient stability chance constraints.

Reinforcement learning (RL), widely applied in microgrid control due to its excellent adaptability to environments and objective functions, commonly utilizes algorithms such as deep Q-network (DQN) [15], deep deterministic policy gradient (DDPG) [16], and proximal policy optimization (PPO) [17]. However, these RL algorithms are not effective in handling hard constraints. In [15], the training is terminated if a hard constraint is violated, and a penalty is added to the reward function when a soft constraint

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is breached. Due to the direct addition of the objective function and constraint penalties as the reward, ensuring adherence to constraints becomes challenging. Therefore, safe RL, which can generate safe policies for both hard and soft constraints to explore within safe zones of the environment, has been developed [18]. Safe RL can generally be categorized into several types: projection-based methods [19], [20], Lagrangian (Lag)-based methods [21], [22], safety layer method [23], among others. Based on constrained policy optimization (CPO) [24], projection-based methods include constrained update projection (CUP) [25], first-order constrained optimization in policy space (FOCOPS) [26], projection-based CPO (PCPO) [27], etc. Lag-based methods include PPO-Lag [28], trust region policy optimization-Lag (TRPO-Lag) [28], constraint-controlled PPO-PID (CPPO-PID) [29], reward CPO (RCPO) [30], etc. Safe RLs are also applied in microgrids, including areas such as energy management [19], [20], voltage control [23], economic dispatch [21], and electric vehicles charging [22].

This paper deals with a more challenging problem as compared to the aforementioned works, i.e., chance-constrained TSEC. Specifically, we develop a safe RL-based, chance-constrained, safety-guaranteed TSEC method for islanded microgrids. The main contributions are summarized as follows:

- 1) We propose a surrogate model enhanced TSEC control, where the deep sigma point processes (DSPP) [31] is used for probabilistic TSE of the islanded microgrid while the safe DRL effectively solves the model while satisfying all hard constraints. DSPP takes post-fault measurement data and emergency control actions as input, outputting the normal distribution of TSE, which is then transformed into a chance constraint for DRL-based control.
- 2) To tackle the challenge of guaranteeing hard constraints, a RCPO-based TSEC method is developed. DSPP, serving as a transient stability chance constraint, is embedded into the islanded microgrid environment. RCPO interacts with the environment in a multi-process parallel manner, significantly accelerating the training process. Results show that the proposed method can efficiently and quickly obtain the optimal TSEC strategy while adhering to all hard constraints.

The rest of the paper is organized as follows. Section II presents the objective and constraints of the traditional islanded microgrid TSEC. Section III shows the proposed DSPP and RCPO-based chance-constrained TSEC framework. Results are presented and analyzed in Section IV, and finally, Section V concludes the paper.

II. PROBLEM STATEMENT

The focus of this study is to find the optimal TSEC strategy for scenarios, where a worst-case three-phase short circuit fault leads to the disconnection of the largest power station in the islanded microgrid. The objective function and constraints are:

A. Objective Function

The objective function aims to minimize load-shedding resulting from the disconnection of the largest power station:

$$\min \sum_{\forall \phi \in \mathcal{P}_n, n \in \mathcal{N}} \Delta P_{\text{Load},n}^{\text{EC},\phi} \quad (1)$$

where $\mathcal{N}$ denotes set of nodes in the microgrid; $\mathcal{P}_n$ denotes the set of phases at node n ; $\Delta P_{\text{Load},n}^{\text{EC},\phi}$ represents the load shed due to emergency control at node n and phase ϕ .

B. Power Flow Equations

For the unbalanced three-phase system, the power flow equations are [32]

$$V_s^\phi (I_s^\phi)^* = P_{G,s}^\phi - P_{D,s}^\phi + j(Q_{G,s}^\phi - Q_{D,s}^\phi), \quad \forall \phi \in \mathcal{P}_s, s \in \mathcal{S} \quad (2)$$

$$V_n^\phi (I_n^\phi)^* = -P_{D,n}^\phi - jQ_{D,n}^\phi + jy_{C,n}^\phi |V_n^\phi|^2, \quad \forall \phi \in \mathcal{P}_n, n \in \mathcal{N} \setminus \mathcal{S} \quad (3)$$

$$I_n^\phi = \sum_{m \in \mathcal{N}_m} \left[\left(\frac{1}{2} \mathbf{Y}_{mn}^{(s)} + \mathbf{Z}_{mn}^{-1} \right) [\mathbf{v}_m]_{\mathcal{P}_{mn}} - \mathbf{Z}_{mn}^{-1} [\mathbf{v}_m]_{\mathcal{P}_{mn}} \right]_{\{\phi\}}, \quad \forall \phi \in \mathcal{P}_n, n \in \mathcal{N} \quad (4)$$

$$P_{G,n}^\phi = P_{\text{DG},n}^\phi + P_{\text{PV},n}^\phi + P_{\text{BESS},n}^{\text{discharge},\phi}, \quad \forall \phi \in \mathcal{P}_n, n \in \mathcal{N} \quad (5)$$

$$P_{D,n}^\phi = P_{\text{BESS},n}^{\text{charge},\phi} + P_{\text{Load},n}^\phi, \quad \forall \phi \in \mathcal{P}_n, n \in \mathcal{N} \quad (6)$$

$$Q_{G,n}^\phi = Q_{\text{DG},n}^\phi + Q_{\text{PV},n}^\phi + Q_{\text{BESS},n}^{\text{discharge},\phi}, \quad \forall \phi \in \mathcal{P}_n, n \in \mathcal{N} \quad (7)$$

$$Q_{D,n}^\phi = Q_{\text{BESS},n}^{\text{charge},\phi} + Q_{\text{Load},n}^\phi, \quad \forall \phi \in \mathcal{P}_n, n \in \mathcal{N} \quad (8)$$

where $\mathcal{S}$, $\mathcal{N}_m$ and $\mathcal{P}_{mn}$ denote set of power supply nodes, neighboring nodes of $m \in \mathcal{N}$, phases of line (m, n) , respectively; V_n^ϕ and I_n^ϕ are complex line-to-ground voltage and complex current injected at phase ϕ of node n , respectively; $P_{G,n}^\phi$ and $P_{D,n}^\phi$ are active power supply and demand at phase ϕ of node n , respectively; $Q_{G,n}^\phi$ and $Q_{D,n}^\phi$ are reactive power supply and demand at phase ϕ of node n , respectively; $\mathbf{Y}_{mn}^{(s)}$ denotes the shunt admittance matrix of line (m, n) ; $\mathbf{Z}_{mn}$ denotes the phase impedance matrix of line (m, n) ; $P_{\text{PV},n}^\phi$, $P_{\text{DG},n}^\phi$, $P_{\text{BESS},n}^{\text{charge},\phi}$, $P_{\text{BESS},n}^{\text{discharge},\phi}$ and $P_{\text{Load},n}^\phi$ denote the active power of PVs, DGs, BESS charge, BESS discharge and loads at phase ϕ of node n , respectively; $Q_{\text{PV},n}^\phi$, $Q_{\text{DG},n}^\phi$, $Q_{\text{BESS},n}^{\text{charge},\phi}$, $Q_{\text{BESS},n}^{\text{discharge},\phi}$ and $Q_{\text{Load},n}^\phi$ denote the reactive power of PVs, DGs, BESS charge, BESS discharge and loads, respectively.

C. DG Constraints

The DG constraints include active and reactive power capacity constraints:

$$\underline{P}_{\text{DG},d} \leq P_{\text{DG},d} \leq \bar{P}_{\text{DG},d}, \quad d \in \mathcal{D} \quad (9)$$

$$\underline{Q}_{\text{DG},d} \leq Q_{\text{DG},d} \leq \bar{Q}_{\text{DG},d}, \quad d \in \mathcal{D} \quad (10)$$

where $\mathcal{D}$ denotes the set of DGs; $\bar{P}_{\text{DG},d}$ and $\underline{P}_{\text{DG},d}$ denote upper and lower limits of $P_{\text{DG},d}$, respectively; $\bar{Q}_{\text{DG},d}$ and

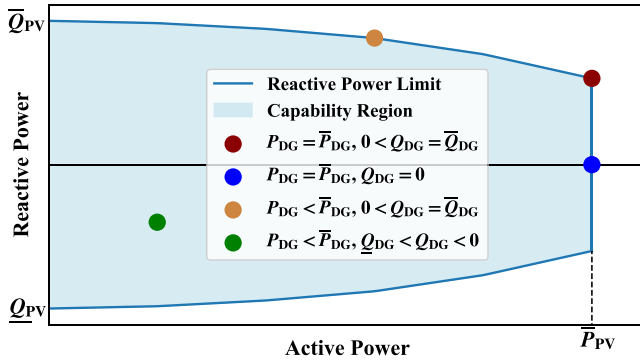


Fig. 1. The capability curve of DGs.

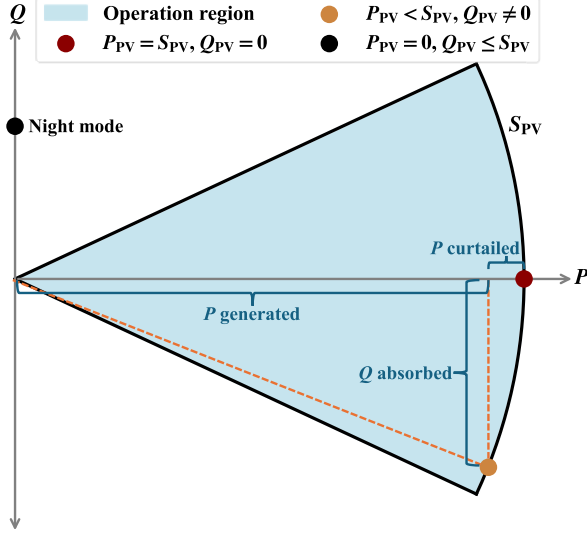


Fig. 2. The operation range of PVs.

$\underline{Q}_{DG,d}$ are upper and lower limits of $Q_{DG,d}$, respectively, following the capability curve shown in Fig. 1.

D. PV Constraints

The PV constraints include active, reactive, and apparent power capacity constraints:

$$0 \leq P_{PV,r} \leq \bar{P}_{PV,r} \leq S_{PV,r}^{\text{rated}}, \quad r \in \mathcal{R} \quad (11)$$

$$\underline{Q}_{PV,r} \leq Q_{PV,r} \leq \bar{Q}_{PV,r}, \quad r \in \mathcal{R} \quad (12)$$

$$\bar{Q}/\underline{Q}_{PV,r} = +/\sqrt{(S_{PV,r}^{\text{rated}})^2 - (P_{PV,r})^2}, \quad r \in \mathcal{R} \quad (13)$$

where $\bar{P}_{PV,r}$ is the maximum value of $P_{PV,r}$ based on the current solar irradiance; $S_{PV,r}^{\text{rated}}$ denotes the rated apparent power of the r -th PV system; $\bar{Q}_{PV,r}$ and $\underline{Q}_{PV,r}$ are upper and lower limits of $Q_{PV,r}$, respectively; $\mathcal{R}$ denotes the set of PVs. The operation range of PV is shown in Fig. 2.

E. BESS Constraints

The BESS constraints include the charging power, discharging power, charging and discharging status, and capacity constraints:

$$P_{\text{BESS},b}^{\text{charge}} \leq P_{\text{BESS},b}^{\text{charge}} \leq \bar{P}_{\text{BESS},b}^{\text{charge}}, \quad b \in \mathcal{B} \quad (14)$$

$$P_{\text{BESS},b}^{\text{discharge}} \leq P_{\text{BESS},b}^{\text{discharge}} \leq \bar{P}_{\text{BESS},b}^{\text{discharge}}, \quad b \in \mathcal{B} \quad (15)$$

$$u_{\text{BESS},b}^{\text{charge}} + u_{\text{BESS},b}^{\text{discharge}} \leq 1, \quad b \in \mathcal{B} \quad (16)$$

$$E_{\text{BESS},b} = E_{\text{BESS},b}^{\text{Init}} + u_{\text{BESS},b}^{\text{charge}} P_{\text{BESS},b}^{\text{charge}} \Delta t - u_{\text{BESS},b}^{\text{discharge}} P_{\text{BESS},b}^{\text{discharge}} \Delta t, \quad b \in \mathcal{B} \quad (17)$$

$$E_{\text{BESS},b} \leq E_{\text{BESS},b} \leq \bar{E}_{\text{BESS},b}, \quad b \in \mathcal{B} \quad (18)$$

where $\mathcal{B}$ denotes the set of BESSs; $\bar{P}_{\text{BESS},b}^{\text{charge}}$ and $P_{\text{BESS},b}^{\text{charge}}$ are upper and lower limits of $P_{\text{BESS},b}^{\text{charge}}$; $\bar{Q}_{\text{BESS},b}^{\text{charge}}$ and $Q_{\text{BESS},b}^{\text{charge}}$ are upper and lower limits of $Q_{\text{BESS},b}^{\text{charge}}$; $u_{\text{BESS},b}^{\text{charge}}$ and $u_{\text{BESS},b}^{\text{discharge}}$ are Boolean variable of charge and discharge states; $E_{\text{BESS},b}$ represents the stored energy of BESS, while $E_{\text{BESS},b}^{\text{Init}}$ denotes its initial value at the current timeslot; Δt is the duration of the action taken by the BESS following TSEC; $\bar{E}_{\text{BESS},b}$ and $E_{\text{BESS},b}$ are upper and lower limits of $E_{\text{BESS},b}$, respectively.

F. Other Static Constraints

Other static constraints include voltage, load-shedding, and topology constraints (no ring networks and islanded):

$$\underline{V}_n \leq V_n^{\phi} \leq \bar{V}_n, \quad \forall \phi \in \mathcal{P}_n, \quad n \in \mathcal{N} \quad (19)$$

$$\underline{P}_{\text{Load},n}^{\phi} \leq \Delta P_{\text{Load},n}^{\text{EC},\phi} \leq P_{\text{Load},n}^{\phi}, \quad \forall \phi \in \mathcal{P}_n, \quad n \in \mathcal{N} \quad (20)$$

$$(N_{\text{Branch}} = N_{\text{Bus}} - 1) \& (N_{\text{Bus},k} \geq 1), \quad k \in \mathcal{K} \quad (21)$$

where $\bar{V}_n$ and $\underline{V}_n$ are upper and lower limits of V_n^{ϕ} ; $\underline{P}_{\text{Load},n}^{\phi}$ is minimum values of $P_{\text{Load},n}^{\phi}$; N_{Branch} , N_{Bus} and $N_{\text{Bus},k}$ denote the number of branches, the number of buses, and the number of branches connected to bus k , respectively; $\mathcal{K}$ denotes the set of buses; $\&$ is the logical conjunction.

G. Transient Stability Constraint

The transient stability is constrained by DAEs [5], [6], and transient stability index (TSI) [33] can be calculated:

$$\dot{\mathbf{x}}_t = \mathbf{f}(\mathbf{x}_t, \mathbf{y}_t, \mathbf{u}, \mathbf{a}, \mathbf{p}) \quad (22)$$

$$\mathbf{0} = \mathbf{g}(\mathbf{x}_t, \mathbf{y}_t, \mathbf{u}, \mathbf{a}, \mathbf{p}) \quad (23)$$

$$\text{TSI}(\mathbf{x}) = \frac{360^\circ - \delta^{\max}}{360^\circ + \delta^{\max}} \times 100 \geq \text{TSI}_B \geq 0 \quad (24)$$

$$\delta^{\max}(\mathbf{x}) = \max_{t \in T_s} (|\delta_{i,t} - \delta_{j,t}|), \quad \forall i, j \in \mathcal{G} \quad (25)$$

where $\mathbf{x}_t$ denotes the state variables, such as the rotor angle δ and rotor speed ω of the synchronous generators, i.e., $\mathbf{x}_t = [\delta_t, \omega_t, \dots]$; $\mathbf{y}_t$ denotes the algebraic variables, such as voltage and current; $\mathbf{u}$ represents the input variables before fault occur; $\mathbf{a}$ is the performed TSEC actions; $\mathbf{p}$ denotes the model parameters; $\mathcal{G}$ denotes the set of all synchronous generators; T_s denotes the whole TDS period; $\delta_{i,t}$ and $\delta_{j,t}$ are the rotor angles of generator i and j at time t ; $\delta^{\max}$ is the maximum rotor angle difference of any two generators. TSI and transient stability are positively correlated. Typically, $\text{TSI}_B = 0$, corresponding to $\delta^{\max} = 360^\circ$, is used as the boundary between transient stable and unstable [34]. For a more conservative result, $\text{TSI}_B = 33.3$, corresponding to $\delta^{\max} = 180^\circ$, is used [35]. However, directly setting a fixed conservative boundary may be overly cautious or still carry certain safety risks. The calculation process from (22) to (25) to obtain TSI is illustrated in Fig. 3.

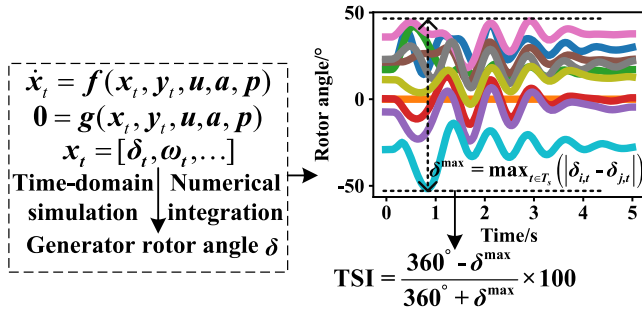


Fig. 3. Calculation process from (22) to (25) to obtain TSI.

III. PROPOSED RCPO-BASED CHANCE-CONSTRAINED SAFETY-GUARANTEED TSEC

A. DSPP for Transient Stability Chance Constraint

GP is a machine learning method achieved through the generalization of the Gaussian probability distribution. GP can quantify predictive uncertainty, outputting the normal distribution of estimations. A GP can be completely defined by its mean function $\mu(X)$ and covariance function $k(X, X')$ [13], [31], [36]:

$$f(X) \sim \mathcal{GP}(\mu(X), k(X, X')) \quad (26)$$

$$\mu(X) = \mathbb{E}[f(X)] \quad (27)$$

$$k(X, X') = \mathbb{E}[(f(X) - \mu(X))(f(X') - \mu(X'))] \quad (28)$$

where $X = \{x_i\}_{i=1}^{N_S}$ denotes the training inputs of GP; N_S is the number of samples; $f(X)$ denotes the GP latent function values. The GP can be calculated analytically, but it becomes computationally prohibitive for large samples, necessitating approximation when N_S is large. For this reason, the inducing point method-based sparse GP (SGP) is developed, which can be applied to very large samples [37]. Deep GP (DGP) is an extension of GP, where a sequence of GP layers forms a hierarchical model [38]. In DGP, the outputs of one GP layer serve as the inputs for the subsequent layer. However, calculating the predicted distribution of DGP is intractable. Therefore, DSPP, a parametric finite mixture method, has been developed to employ sigma point approximation to simplify this process. For a detailed account of the development and derivation process from GP to DSPP, refer to [13], [31].

In this paper, DSPP is used for probabilistic TSE. The input matrix of DSPP is $\mathcal{I} = [\mathbf{P}_{PV}^{\text{FC}}, \mathbf{P}_{DG}^{\text{FC}}, \mathbf{P}_{\text{BESS}}^{\text{FC}}, \mathbf{P}_{\text{Load}}^{\text{FC}}, \mathbf{Q}_{\text{Load}}^{\text{FC}}, \mathbf{E}_{\text{BESS}}^{\text{FC}}, \mathbf{U}_{\text{Switch}}^{\text{FC}}, \Delta \mathbf{P}_{\text{DG}}^{\text{EC}}, \Delta \mathbf{P}_{\text{BESS}}^{\text{EC}}, \Delta \mathbf{P}_{\text{Load}}^{\text{EC}}, \mathbf{U}_{\text{Switch}}^{\text{EC}}]^T$, where $\mathbf{U}_{\text{Switch}}^{\text{EC}}$ denotes the Boolean value indicating the on or off state of switches; the superscript FC indicates the measured values through smart meters after fault clearance; the superscript EC indicates the executed TSEC actions. The output variables of DSPP are the mean of TSI μ_{TSI} and the standard deviation of TSI σ_{TSI} . DSPP is used as a surrogate model for transient stability constraint (22) to (25), accelerating TSE. Based on the cumulative distribution function (CDF) of the normal distribution, the value of Λ that ensures the probability of the predicted TSI being within the range $[\mu_{\text{TSI}} - \Lambda\sigma_{\text{TSI}}, \mu_{\text{TSI}} + \Lambda\sigma_{\text{TSI}}]$

meets a predefined confidence level ε can be calculated by

$$\frac{1}{\sqrt{2\pi}\sigma_{\text{TSI}}} \int_{\mu_{\text{TSI}} - \Lambda\sigma_{\text{TSI}}}^{+\infty} \exp\left(-\frac{(\text{TSI}^{\text{TSE}} - \mu_{\text{TSI}})^2}{2\sigma_{\text{TSI}}^2}\right) d\text{TSI}^{\text{TSE}} \geq \varepsilon \quad (29)$$

where TSI^{TSE} denotes the predicted TSI. The DSPP-based transient stability chance constraint is rewritten as:

$$\mathbb{P}(\text{TSI}^{\text{TSE}} \geq \mu_{\text{TSI}} - \Lambda\sigma_{\text{TSI}}) \geq \varepsilon \quad (30)$$

where $\mathbb{P}$ denotes the probability of satisfying the constraint.

B. Constrained Markov Decision Processes

The MDP is defined by a tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, R, \rho_0, \gamma)$, where $\mathcal{S}$ is the state space; $\mathcal{A}$ is the control space; $\mathcal{P} : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow [0, 1]$ is the transition probability matrix; $R : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}$ is the reward function; $\rho_0 : \mathcal{S} \rightarrow [0, 1]$ is the starting state distribution; $\gamma \in [0, 1]$ is the discount factor for future reward. The value of each state s following policy π is

$$V_R^\pi(s) = \mathbb{E}_{\tau \sim \pi} \left[\sum_t \gamma^t r(s_t, a_t, s_{t+1}) | s_0 = s \right] \quad (31)$$

where $\tau \sim \pi$ denotes that the distribution over trajectories depends on π : $s_0 \sim \rho_0$, $a_t \sim \pi(\cdot | s_t)$, $s_{t+1} \sim \mathcal{P}(\cdot | s_t, a_t)$.

The goal of MDP is to learn a policy π that maximizes reward performance $\mathcal{J}_R^\pi$, which is the discounted sum of rewards [39], [40], [41]

$$\mathcal{J}_R^\pi = \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t, s_{t+1}) \right] \quad (32)$$

where $\mathbb{E}$ is the expectation function.

The CMDP $\mathcal{M}_C = (\mathcal{S}, \mathcal{A}, \mathcal{P}, R, \rho_0, \gamma, C)$ is an extension of a standard MDP $\mathcal{M}$ with a constraint function $C(s)$ and a threshold α . The constraint can be a discounted sum, the average sum, and others. The general form of expectation of constraint $\mathcal{J}_C^\pi$ is

$$\mathcal{J}_C^\pi = \mathbb{E}_{\tau \sim \pi} [C(s)] \quad (33)$$

The problem becomes

$$\max_{\pi \in \Pi} \mathcal{J}_R^\pi, \quad \text{s.t. } \mathcal{J}_C^\pi \leq \alpha \quad (34)$$

C. Reward Constrained Policy Optimization

A common method for solving CMDP is Lagrangian relaxation, which transforms constrained problems into equivalent unconstrained problems [18]. In this unconstrained problem, in addition to optimizing the objective function, constraints are treated as penalty terms added to the reward.

$$\begin{aligned} \min_{\lambda \geq 0} \max_{\theta} \mathcal{L}(\lambda, \theta) \\ = \min_{\lambda \geq 0} \max_{\theta} [\mathcal{J}_R^{\pi_\theta} - \lambda \cdot (\mathcal{J}_C^{\pi_\theta} - \alpha)] \end{aligned} \quad (35)$$

where θ is the parameters of the policy, and a parameterized policy represented as π_θ ; $\mathcal{L}$ denotes the Lagrangian; λ is the Lagrange multiplier. As λ increases, the solution to (35) converges to that of (34), indicating a two-timescale approach. On the faster timescale, θ is determined by solving (35), and

on the slower timescale, λ is increased until the constraint is fully satisfied. In CPO, the λ and θ are updated through

$$\lambda_{k+1} = \Gamma_\lambda[\lambda_k - \eta_1(k) \nabla_\lambda \mathcal{L}(\lambda_k, \theta_k)] \quad (36)$$

$$\theta_{k+1} = \Gamma_\theta[\theta_k + \eta_2(k) \nabla_\theta \mathcal{L}(\lambda_k, \theta_k)] \quad (37)$$

where Γ is the projection operator, it projects θ onto a compact and convex set to maintain the stability of the iteration and projects λ into the range $[0, \bar{\lambda}]$. $\eta_1(k)$ and $\eta_2(k)$ are step-sizes, and it is necessary to ensure the policy update is performed on a faster timescale than that of the constraint cost coefficient λ . $\nabla_\theta \mathcal{L}$ and $\nabla_\lambda \mathcal{L}$ are derived from equation (35), where the formulation for $\nabla_\theta \mathcal{L}$ is derived using the log-likelihood trick.

$$\nabla_\lambda \mathcal{L}(\lambda, \theta) = -(\mathbb{E}_{\tau \sim \pi_\theta}[C(s)] - \alpha) \quad (38)$$

$$\nabla_\theta \mathcal{L}(\lambda, \theta) = \nabla_\theta \mathbb{E}_{\tau \sim \pi_\theta}[\log \pi(s, a; \theta)[R(s) - \lambda \cdot C(s)]] \quad (39)$$

Based on the assumptions about the values of η_1 and η_2 , along with a standard stability assumption for the iterates and bounded noise, it can be proven that the iterates $(\theta_n; \lambda_n)$ converge to a fixed point (a local minimum) almost surely. Furthermore, under the assumptions that the value $V_R^\pi(s)$ is bounded for all policies and that every local minimum of $J_C^\pi(\theta)$ is a feasible solution, it can be proven that this fixed point is a feasible solution [30].

Recently, Actor-Critic-based methods have seen significant development, where the actor learns a policy π , while the critic learns the value function using temporal-difference learning - the recursive Bellman equation. However, training a critic requires a recursive property, which is not applicable to general constraints. RCPO overcomes this issue by training the actor and critic using an alternative guiding penalty - the discounted penalty. The value of the discounted (guiding) penalty is

$$V_{C_Y}^\pi(s) \triangleq \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^{\infty} \gamma^t c(s_t, a_t) \mid s_0 = s \right] \quad (40)$$

The penalized reward functions are

$$\hat{r}(\lambda, s, a) \triangleq r(s, a) - \lambda c(s, a) \quad (41)$$

$$\begin{aligned} \hat{V}^\pi(\lambda, s) &\triangleq \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^{\infty} \gamma^t \hat{r}(\lambda, s_t, a_t) \mid s_0 = s \right] \\ &= \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^{\infty} \gamma^t (r(s_t, a_t) - \lambda c(s_t, a_t)) \mid s_0 = s \right] \\ &= V_R^\pi(s) - \lambda V_{C_Y}^\pi(s) \end{aligned} \quad (42)$$

Unlike the original Lagrangian relaxation (35), for fixed π and λ , the penalized value (42) can be evaluated by a temporal difference (TD)-learning critic. Then, actor and critic can be updated following (42), and λ is updated following (36). Based on this, RCPO is proposed, which employs a three-timescale approach. On the fast timescale, an alternative, discounted objective is estimated using a TD-critic. On the intermediate timescale, the policy is learned using policy gradient methods. And on the slow timescale, the constraint cost coefficient λ is learned by ascending on the original constraint [30].

D. Proposed RCPO-Based TSEC Framework

Based on the DSPP-based transient stability chance constraint and RCPO, a TSEC framework is proposed. In this framework, DSPP replaces the time-consuming TDS as part of the microgrid environment for TSE, effectively quantifying uncertainties. Meanwhile, RCPO interacts with the DSPP-based environment to determine the emergency control policy, while ensuring compliance with the power system's constraints. The interaction between RCPO and the environment is based on state s , action a , reward r , constraint cost c and terminated condition, which are defined in detail as follows:

A) State

The state of the microgrid, determined by data measured by smart meters following fault clearance, can be defined as:

$$s = [\mathbf{P}_{PV}^{FC}, \mathbf{P}_{DG}^{FC}, \mathbf{P}_{BESS}^{FC}, \mathbf{P}_{Load}^{FC}, \mathbf{Q}_{Load}^{FC}, \mathbf{E}_{BESS}^{FC}, \mathbf{U}_{Switch}^{FC}]^T \quad (43)$$

B) Action

The action of the TSEC includes the DGs dispatch, BESSs control, load-shedding, and topology transformation, which can be defined as:

$$a = [\Delta \mathbf{P}_{DG}^{EC}, \Delta \mathbf{P}_{BESS}^{EC}, \Delta \mathbf{P}_{Load}^{EC}, \mathbf{U}_{Switch}^{EC}]^T \quad (44)$$

The action a cannot take arbitrary values as it includes several hard constraints that must be satisfied, such as the upper and lower bounds of the action, load-shedding constraints, and topology constraints. Therefore, the following considerations or treatments are necessary:

- 1) The active and reactive loads are shed in proportion, hence only $\Delta \mathbf{P}_{Load}^{EC}$ is required.
- 2) In the battery capacity constraints from (16) to (18), two Boolean variables of charge and discharge states are introduced, which constitutes a hard constraint. Including it directly in the constraint cost of RCPO would increase complexity. Therefore, $\mathbf{E}_{BESS}^{FC}$ is set to positive values for discharging and negative values for charging, ensuring that the BESS's charging and discharging state constraints are always satisfied.
- 3) The upper and lower limits of all actions are restricted by the action space's value range, ensuring constraint satisfaction.
- 4) The constraints (10) and (12) are satisfied when automatic voltage regulation is enabled, which automatically adjusts reactive power output to maintain the generator terminal voltage near its setpoint, following the capability curve in Fig. 1 and the operation range in Fig. 2, respectively.
- 5) $\Delta \mathbf{P}_{Load}^{EC}$ is randomly generated, but when actually shedding the load, it will be checked against the current load $\mathbf{P}_{Load}$. Any excess amount will be shielded, ensuring that $\Delta \mathbf{P}_{Load}^{EC} \leq \mathbf{P}_{Load}$.
- 6) Similar to $\Delta \mathbf{P}_{Load}^{EC}$, the BESS capacity constraint (18) can also be met by shielding the excessive charging or discharging power $\Delta \mathbf{P}_{BESS}^{EC}$ within the feasible range.
- 7) $\mathbf{U}_{Switch}^{EC}$ represents the switch state Boolean variables, indicating on or off states. Since the actor's output is continuous, these variables are normalized to 0 or 1 through softmax activation to match the binary nature of

the switch states. Additionally, the number of topologies corresponding to switch states is a power of 2. When the number of switches is large, the proportion of feasible topologies to the total number of topologies is very low, greatly increasing the difficulty of imposing constraints. Thus, binary encoding is applied to the topologies, and infeasible topologies are projected onto feasible ones through projection. For example, in the numerical analysis of this paper, the number of switches in the microgrid is 13, resulting in 189 feasible topologies. The proportion of feasible topologies to the total number of topologies is $189/2^{13} = 2.3\%$, which is very low. Directly adding this to the constraint cost makes it difficult to converge. Therefore, by using an 8-bit binary encoding, the proportion of feasible topologies to the total increases to $189/2^8 = 73.8\%$. The remaining 67 infeasible topologies are directly projected onto a preset, corresponding feasible topology.

C) Reward

The objective function (1) of the proposed method aims to minimize the total power of load-shedding. Since the reward is maximized, the reward is formulated as:

$$\mathbf{r} = - \sum_{l \in \mathcal{D}} \Delta P_{\text{Load},l}^{\text{EC}} \quad (45)$$

It should be noted that here, $\Delta P_{\text{Load}}^{\text{EC}}$ is directly output by the actor, not after shielding, to intrinsically reduce the total amount of load-shedding.

D) Constraint Cost

Voltage and transient stability are nonlinear, non-convex functions of the action, and cannot be simply projected or shielded to restrict them within feasible ranges. Therefore, voltage constraints and DSPP-based transient stability chance constraints are added to the constraint cost as follows:

$$\mathbf{c} = \beta_V c_V + \beta_{\text{TSI}} c_{\text{TSI}} \quad (46)$$

where β_V and β_{TSI} are the weights for the two penalties, respectively;

$$c_V = \sum_{n \in \mathcal{N}} c_{V,n} \quad (47)$$

$$c_{V,n} = \begin{cases} \sum (V_n - \underline{V}_n), V_n < \underline{V}_n \\ \sum (\bar{V}_n - V_n), V_n > \bar{V}_n, n \in \mathcal{N} \\ 0, \underline{V}_n \leq V_n \leq \bar{V}_n \end{cases} \quad (48)$$

$$c_{\text{TSI}} = \begin{cases} \Lambda \sigma_{\text{TSI}} - \mu_{\text{TSI}}, \mu_{\text{TSI}} < \Lambda \sigma_{\text{TSI}} \\ 0, \mu_{\text{TSI}} \geq \Lambda \sigma_{\text{TSI}} \end{cases} \quad (49)$$

Both (48) and (49) impose penalties only in cases of constraint violation, meaning $\mathbf{c} = 0$ when all constraints are satisfied.

E) Termination Condition

Since both the voltage stability constraint and the transient stability constraint are hard constraints, the termination condition is set to $\mathbf{c} = 0$.

In summary, the process of RCPO interacting with the DSPP-based microgrid environment to update parameters is illustrated in the Algorithm III-D, and the overall framework is depicted in Fig. 4.

Algorithm 1 RCPO-Based TSEC

Input: learning rates $\eta_1(k) < \eta_2(k) < \eta_3(k)$, threshold $\alpha = 0$, parameters of microgrid

Output: policy parameters θ

- 1: Initialize actor parameters $\theta = \theta_0$, critic parameters $v = v_0$, Lagrange multipliers $\lambda = 0$
- 2: **for** $k = 0, 1, \dots$ **do**
- 3: Initialize state $s_0 \sim \rho_0$
- 4: **for** $t = 0, 1, \dots, T-1$ **do**
- 5: In a multi-process parallel approach: sample action $a_t \sim \pi$, perform a_t within the DSPP-based microgrid environment, receive reward r_t and penalties c_t , then relay this information back to RCPO
- 6: (41) and (42) $\rightarrow$ **Penalized reward update:** $\hat{R}_t = r_t - \lambda c_t + \gamma \hat{V}(\lambda, s_t; v_k)$
- 7: (41) and (42) $\rightarrow$ **Critic update:** $v_{k+1} \leftarrow v_k - \eta_3(k) \left[\partial \left(\hat{R}_t - \hat{V}(\lambda, s_t; v_k) \right)^2 / \partial v_k \right]$
- 8: (37) and (39) $\rightarrow$ **Actor update:** $\theta_{k+1} \leftarrow \Gamma_\theta \left[\theta_k + \eta_2(k) \nabla_\theta \hat{V}(\lambda, s) \right]$
- 9: **end**
- 10: (36) and (38) $\rightarrow$ **Lagrange multipliers update:** $\lambda_{k+1} \leftarrow \Gamma_\lambda \left[\lambda_k + \eta_1(k) (J_C^{\pi_\theta} - \alpha) \right]$
- 11: **end**
- 12: **Online:** policy parameters θ are used to generate TSEC action $\mathbf{a}$ (44)

IV. TEST RESULTS

The proposed method is tested on a real 52-bus microgrid, which has 1 power station (composed of 2 NGs and 2 DGs), 26 DGs, 10 PVs, 1 BESS, 27 loads, and 13 switches, corresponding to 189 feasible topologies. The power station is located at the forefront of the microgrid to supply power and represents the microgrid's largest power supply facility. When a three-phase short-circuit fault occurs on the transmission line connected to this power station, the circuit breaker trips to isolate the fault, simultaneously disconnecting the power station and resulting in a significant loss of system power, necessitating TSEC. The fault duration is 100ms, and the TSEC is executed 50ms after the fault clearance. The structure of the microgrid is shown in Fig. 5.

A. DSPP Training and Performance Evaluation

Based on the actual measured data for each PV, DG, BESS, load, and switch, set their value ranges to generate samples for training the DSPP. DSPP is constructed using GPyTorch, a Gaussian process library implemented with PyTorch [42]. The configuration consists of a minibatch size of 500, 5 quadrature sites, 2 GPs in the hidden layer, and 500 training epochs. The total sample size is 20,000, with 80% used for training and 20% for testing. It also includes 300 inducing points in each hidden layer and employs the Matérn kernel. Optimization is performed using the Adam optimizer with an initial learning rate of 0.01. The performance of DSPP is estimated by transient stable and unstable classification accuracy, mean absolute error (MAE) between TSI target and

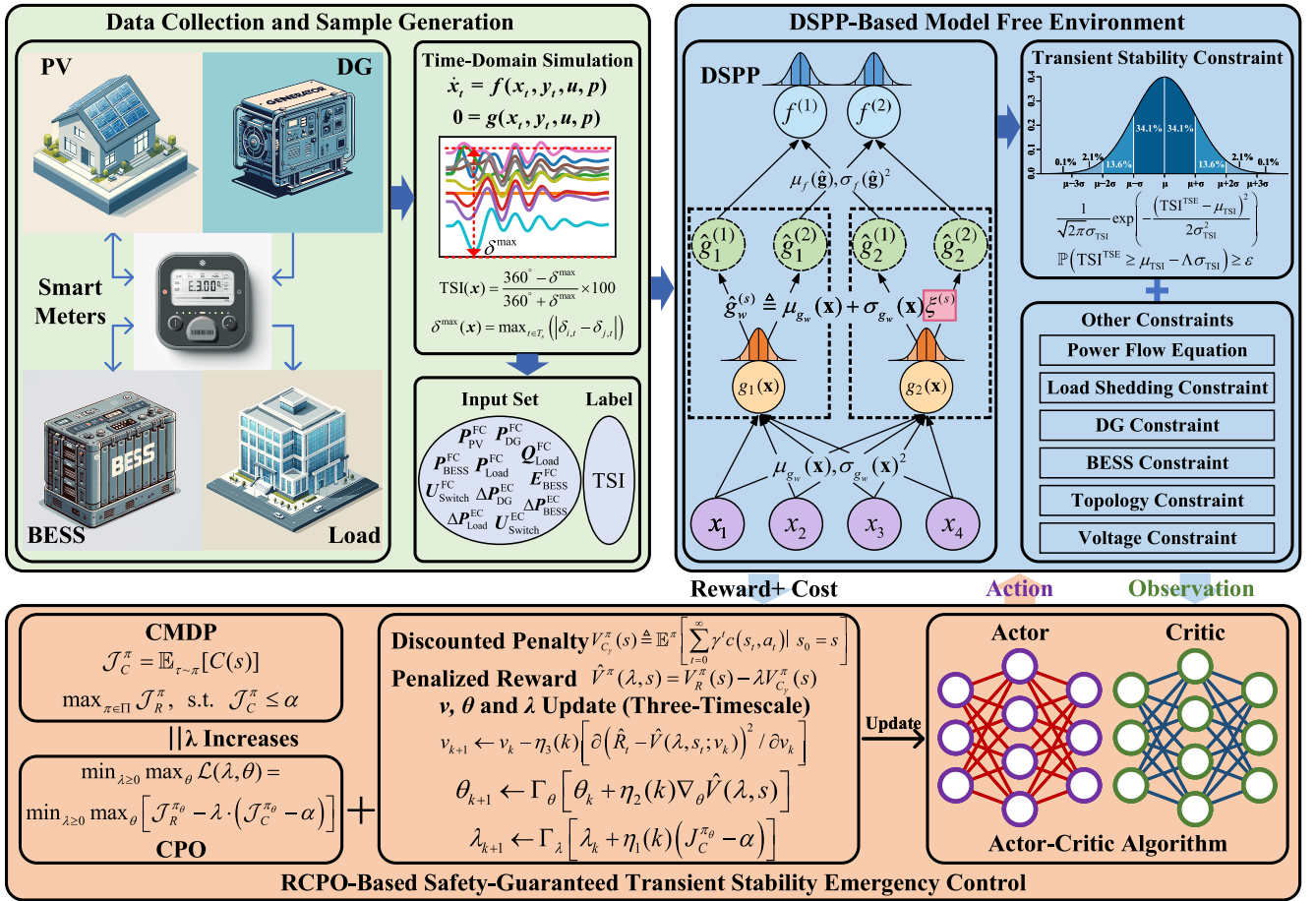


Fig. 4. Proposed DSPP and RCPO-Based Chance-Constrained Safety-Guaranteed TSEC Framework.

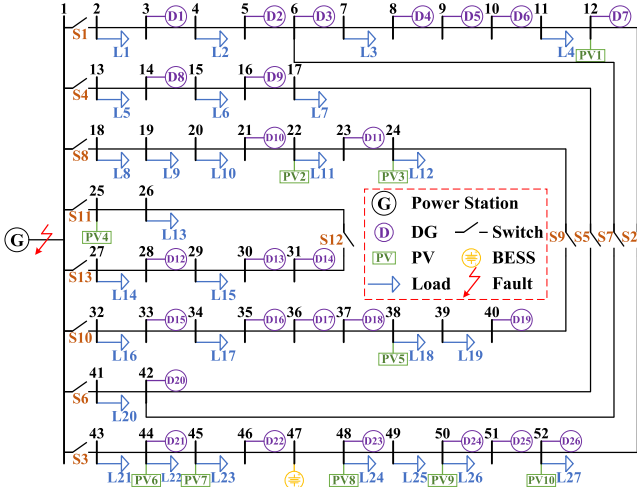


Fig. 5. One-line diagram of the microgrid.

μ_{TSI} , false alarm rate (FA), and miss detection rate (MD). The equations of classification accuracy, MAE, FA, and MD are

$$A_P = \frac{1}{N_T} \sum_{i=1}^{N_T} (\mu_{TSI,i} \times TSI_{T,i} > 0) \quad (50)$$

$$MAE = \frac{1}{N_T} \sum_{i=1}^{N_T} |\mu_{TSI,i} - TSI_{T,i}| \quad (51)$$

$$FA = \frac{1}{N_T} \sum_{i=1}^{N_T} (\mu_{TSI,i} < 0 \text{ and } TSI_{T,i} > 0) \quad (52)$$

$$MD = \frac{1}{N_T} \sum_{i=1}^{N_T} (\mu_{TSI,i} > 0 \text{ and } TSI_{T,i} < 0) \quad (53)$$

where A_P is the classification accuracy; N_T is the number of test set samples; $TSI_{T,i}$ denotes the i -th TSI target of the test set.

The performance of DSPP is also compared with several other probabilistic and deterministic estimation models. For probabilistic estimation models, the GP, SGP, DGP, and deep kernel learning (DKL) [43] are compared. For deterministic estimation models, the four-layer stacked autoencoder (SAE), the three-layer artificial neural network (ANN), and the four-layer DBN are compared. The hyperparameters of the above models are tuned to have optimal performances. The performance comparison results of the above models are shown in Table I. It is observed that the classification accuracy of the GP model is only 25.45%. This low accuracy can be attributed to the limitations of GP in handling large-scale and complex problems, resulting in convergence issues. Among all probabilistic models, it is evident that the DSPP utilized

TABLE I
COMPARISON OF PERFORMANCE OF DIFFERENT ESTIMATION MODELS

Type	Model	A_P	MAE	FA	MD
Probabilistic model	DSPP	99.60%	1.23	0.25%	0.15%
	GP	25.45%	108.3	74.55%	0.00%
	SGP	97.70%	20.8	0.40%	1.90%
	DGP	99.15%	4.70	0.05%	0.80%
	DKL	99.50%	4.60	0.30%	0.20%
Deterministic model	SAE	99.25%	2.38	0.25%	0.50%
	ANN	99.40%	1.86	0.30%	0.30%
	DBN	99.50%	2.66	0.30%	0.20%

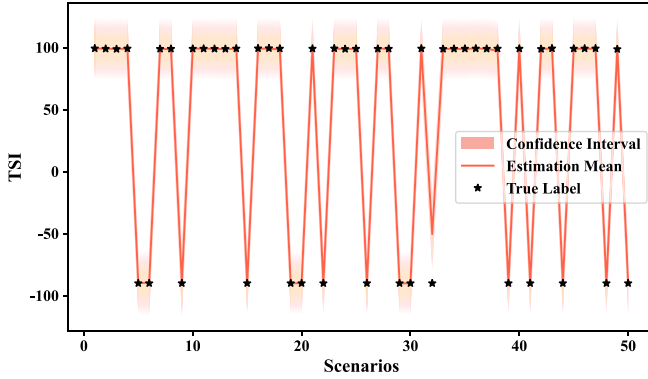


Fig. 6. The estimation performance of DSPP for 50 random scenarios.

in this study outperforms other models in terms of A_P and MAE. Additionally, DSPP exhibits the lowest MD, which is crucial for the reliability of the TSEC, as FA increases the number of control actions, while MD may cause the controller to remain inactive in unsafe scenarios. Furthermore, compared to deterministic models, the DSPP still maintains advantages in terms of A_P and MAE, demonstrating that DSPP not only possesses the highest estimation accuracy but also can quantify estimation uncertainty.

To demonstrate the estimation performance of DSPP, 50 scenarios are randomly selected, encompassing both stable and unstable scenarios. They are estimated using DSPP and compared with the true labels, as shown in Fig. 6. Additionally, based on the estimation mean μ_{TSI} , a confidence interval $[\mu_{TSI} - 3\sigma_{TSI}, \mu_{TSI} + 3\sigma_{TSI}]$ was plotted, within which the probability of the TSI^{TSE} falling is 99.74%. From Fig. 6, it can be observed that, except for the 32-nd scenario, the μ_{TSI} closely match the true labels, consistent with an MAE of 1.23. It should be noted that although the true label of the 32-nd scenario does not fall within the confidence interval, the μ_{TSI} is approximately -50 , indicating that TSEC is still required. Therefore, this does not affect the values of A_P , FA, and MD. The DSPP-based TSI evaluation takes approximately 10ms, achieving a $1,000\times$ speedup compared to traditional TDS, making it suitable for real-time emergency control requirements.

B. RCPO-Based Safety-Guaranteed TSEC

According to Section III-D, RCPO is utilized to execute TSEC for the microgrid in response to a power station disconnect caused by a three-phase short circuit. The training uses 16 parallel environments to accelerate the process. The total

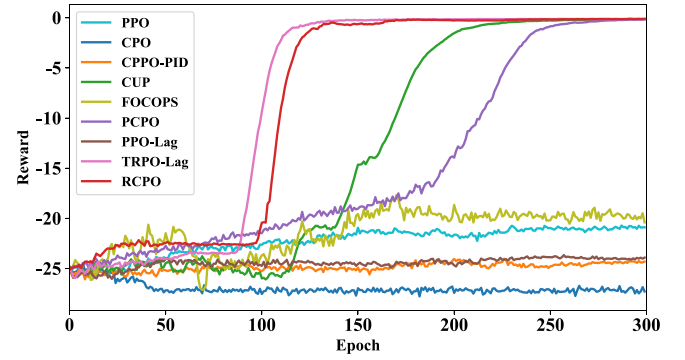


Fig. 7. Comparison of reward trajectories of different RL models during the training process.

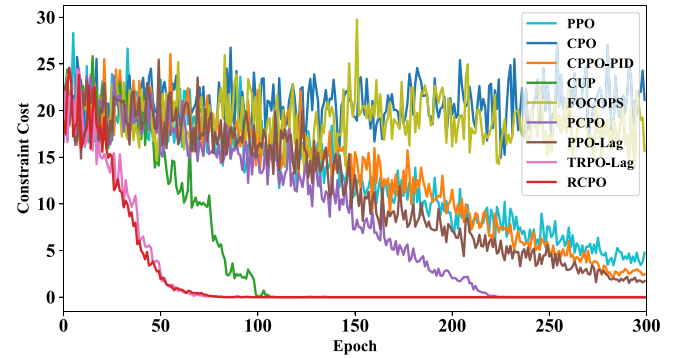


Fig. 8. Comparison of constraint cost trajectories of different RL models during the training process.

training steps amount to 6,000,000, with each environment executing 20,000 steps per policy iteration. The Lagrangian multiplier is initially set at 0.1, with a learning rate of 0.035. It should be noted that, due to the presence of numerous DGs and NGs in this microgrid, which provide voltage support, the voltage always remains within a safe and stable range. Therefore, there is no issue with voltage safety, and only transient stability constraints are considered. In addition to RCPO, a PPO with a constant constraint cost coefficient and other safe RL models are also trained to compare performance. The other safe RL models include CPO, CPPO-PID, CUP, FOCOPS, PCPO, PPO-Lag, and TRPO-Lag. The rewards and penalties of constraints during the training process for all RL models are shown in Fig. 7 and Fig. 8, respectively. Fig. 7 shows that only TRPO-Lag, RCPO, CUP, and PCPO converge close to 0, while the other five RL models remain in an oscillating state. Among the four convergent safe RL models, TRPO-Lag converges the fastest, with RCPO very close behind, and the other two converging more slowly. Similarly, from Fig. 8, it is observed that the same four safe RL models converge to 0, while the other five models do not converge. The convergence speed of RCPO and TRPO-Lag is completely identical, while CUP and PCPO converge more slowly.

To more finely compare the performance of different RL models, the rewards and penalties of each model at the end of training are compared. Additionally, 30 new scenarios are randomly generated, and the trained RL models are tested on these scenarios. The results are presented in Table II. For the

TABLE II
COMPARISON OF PERFORMANCE OF DIFFERENT RL MODELS

Type	Model	Training		Testing	
		Reward	Cost	Reward	Cost
RL	PPO	-25.65	4.76	-93.25	52.56
	CPO	-27.29	21.15	-27.09	24.67
	CPPO-PID	-24.30	2.44	-19.52	23.77
	CUP	-0.11	0.01	-33.19	31.79
Safe RL	FOCOPS	-20.40	15.67	-18.03	19.66
	PCPO	-0.15	0.00	-19.03	0.00
	PPO-Lag	-23.91	1.73	-22.05	24.38
	TRPO-Lag	-0.12	0.01	-2.91	0.00
	RCPO	-0.09	0.00	-0.53	0.00

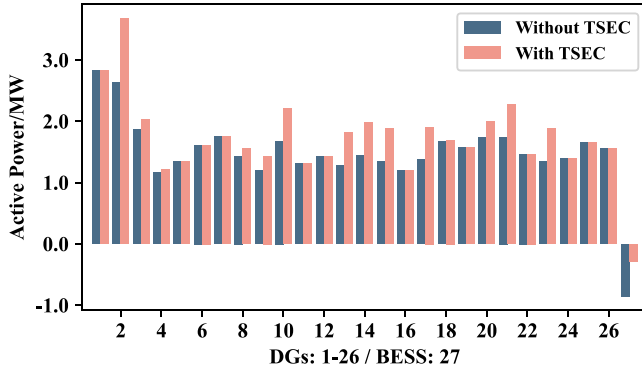


Fig. 9. Comparison of the active power output of DGs and BESS without and with TSEC.

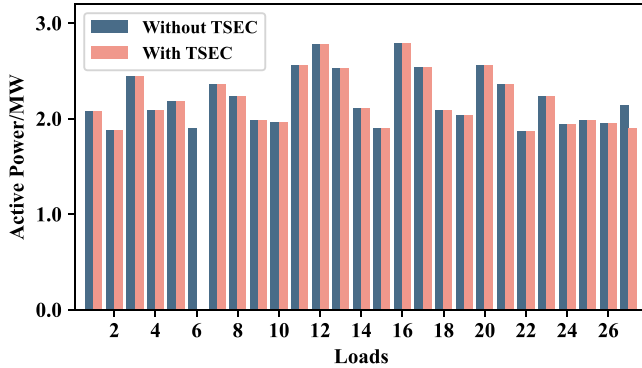


Fig. 10. Comparison of active power demand of loads without and with TSEC.

training stage, consistent with Fig. 7 and Fig. 8, CUP, PCPO, TRPO-Lag, and RCPO's rewards and penalties approach 0. Among them, RCPO exhibits the best final performance with a reward of only -0.09 and a constraint cost of 0.00. This indicates that in the parallel 16 environments, RCPO consistently identified TSEC strategies that satisfy constraints with the minimum total load-shedding required. For the training stage, only PCPO, TRPO-Lag, and RCPO can satisfy constraints across all 16 parallel environments, with RCPO achieving the highest reward of -0.53. Based on the above analysis, it is evident that the RCPO demonstrates the best performance among all the RL models.

Furthermore, for a random transient unstable scenario, RCPO is used to execute TSEC. The comparison of DGs and BESS without and with TSEC is shown in Fig. 9, while the results of load-shedding are depicted in Fig. 10.

Due to the power station outage, the DGs and BESS needed to increase their active power output, while loads had to be shed to deal with the power loss and potential transient instability. As can be seen from Fig. 9, 14 DGs increased power output, including DGs 2, 3, 4, 8, 9, 10, 13, 14, 15, 17, 18, 20, 21 and 23. Meanwhile, the BESS reduced the system's total load by decreasing its charging power. Due to the dispatch of DGs and BESS, the total generation increase is 6.22MW, with DGs contributing 5.66MW and BESS contributing 0.56MW. Similarly, Fig. 10 displays the results of load-shedding. However, given that the reward function aims to minimize total load-shedding, RCPO prioritizes adjusting DGs, BESS, and the state of switches to enhance transient stability. Consequently, only two loads are reduced: load 6 is cut to 0, reducing 1.90MW, and load 27 is decreased by 0.24MW. The topology transformation of the microgrid is shown in Fig. 11, where the red lines indicate the closing of switches. The on and off states of switches 3, 4, 6, 7, 8, 9, 11, and 12 are reversed.

Finally, to compare the transient stability of microgrid without and with TSEC, Fig. 12 and Fig. 13 respectively illustrate the rotor angle trajectories of DGs following the fault without any action and after implementing the control strategy obtained from RCPO. In Fig. 12, without executing any action, the rotor angle trajectories of DGs diverge, indicating transient instability of the microgrid, necessitating the implementation of TSEC, with a TSI of -99.23. Conversely, in Fig. 13, by executing the emergency control strategy, the rotor angle trajectories of DGs converge after minor oscillations, with a TSI of 97.36, demonstrating the reliability of the proposed method. It should be noted that since the rotor angle trajectories in Fig. 13 converge completely to near zero between 1s and 10s, only the first second is depicted to highlight the details after the fault.

C. Comparison With Other TSEC Methods

In this section, for the scenario described in Section IV-B, the proposed method is compared with other RL methods, sensitivity-based methods (SM), and EA in terms of performance regarding reward, constraint cost, and computational time. RL and safe RL methods include PPO, CPO, CPPO-PID, CUP, FOCOPS, PCPO, PPO-Lag, and TRPO-Lag. The sensitivity-based method is widely used in traditional TSEC, utilizing the sensitivity of adjustable devices to the system's transient stability (e.g., TSI). Through sensitivity analysis, the sensitivities of DGs and loads are calculated to identify which equipment needs adjustment and the amount of power adjustment required. Mixed integer linear programming (MILP) is employed to solve this problem, taking into account linearized power flow (LPF) and topology transformation. This can be expressed as:

$$\min \sum_{l \in \mathcal{D}} \Delta P_{\text{Load},l}^{\text{EC}} \quad (54)$$

$$\text{s.t. LPF, (5)~(21)} \quad (55)$$

$$\text{TSI}_0 + \sum_{i \in \mathcal{A}} J_i \Delta a_i > 33.3 \mid \mathbf{J} = \frac{\partial \text{TSI}}{\partial \mathbf{a}} \quad (56)$$

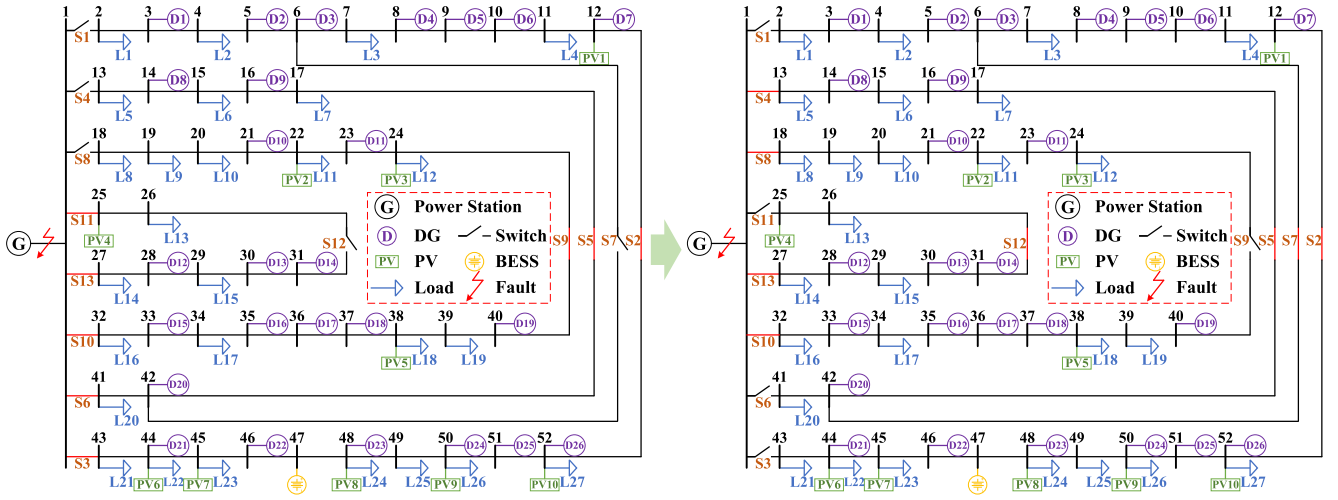


Fig. 11. Topology transformation corresponding to TSEC.

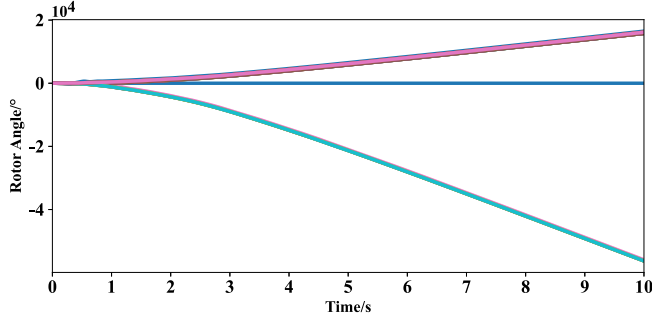


Fig. 12. Rotor angles of all DGs without the TSEC.

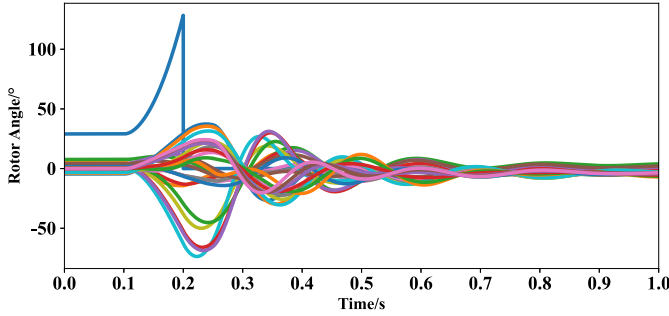


Fig. 13. Rotor angles of all DGs with the TSEC.

where TSI_0 is the TSI without any TSEC action; J is the Jacobian matrix; the TSI after TSEC being greater than 33.3 is equivalent to $\delta^{\max}$ being less than 180° . Since topology reconfiguration is required, we also compared the proposed method with the widely used NSGA-II algorithm due to its flexibility in adjusting the on or off state of switches [10], [44], [45]. To accelerate NSGA-II, we utilized an ANN for enhancement, which can be expressed as:

$$\min \sum_{l \in \mathcal{D}} \Delta P_{Load,l}^{EC} \quad (57)$$

$$\text{s.t. } (2) \sim (21) \quad (58)$$

$$TSI_{ANN}^{TSE}(\mathbf{a}) > 33.3 \quad (59)$$

TABLE III
COMPARISON OF PERFORMANCE OF DIFFERENT TSEC METHODS

Type	Model	Reward	Cost	Calculation time
SM	MILP	-10.51	67.46	>5min
EA	NSGA-II	-6.37	0.00	33s
RL	PPO	-31.08	12.51	
	CPO	-22.34	2.82	
	CPPO-PID	-27.95	2.44	
	CUP	-9.55	50.76	
Safe RL	FOCOPS	-38.84	61.57	<20ms
	PCPO	-10.82	0.00	
	PPO-Lag	-22.09	14.68	
	TRPO-Lag	-5.09	0.00	
	RCPO	-2.14	0.00	

The comparison results are shown in Table III. Table III shows that only NSGA-II, PCPO, TRPO-Lag, and RCPO can satisfy the transient stability constraint, among which RCPO has the highest reward, i.e., the least total load-shedding. For NSGA-II, even with acceleration using ANN, it remains highly time-consuming and fails to meet the requirements of TSEC. The MILP method calculates the sensitivity only once through TDS and does not update it during the TSEC iteration process. Therefore, the constant sensitivity becomes unreliable during iteration, leading to convergence and control failure. Additionally, it should be noted that the calculation time required for a single sensitivity calculation is more than 5 min, while all RL methods can obtain control strategies within 20ms, meeting the requirements for emergency control, such as the 50ms emergency control time in this paper.

D. Impacts of Dynamic Loads and Low-Voltage Ride-Through

Low voltage ride-through (LVRT) enables generators and distributed energy resources to remain connected during short-term voltage dips, preventing large-scale disconnection due to transient faults. This functionality is implemented through an LVRT relay, which disconnects the generator if the low

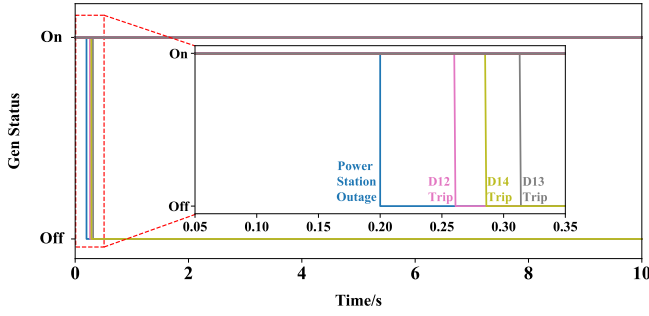


Fig. 14. Generator status trajectories after setting the LVRT relay.

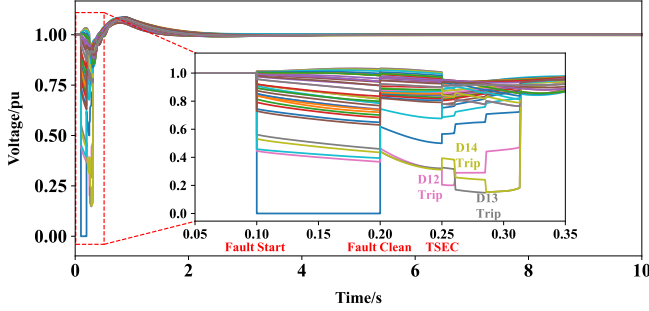


Fig. 15. Generator voltage trajectories after setting the LVRT relay.

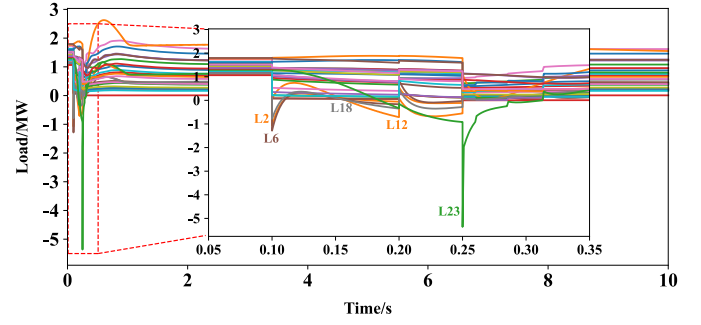


Fig. 16. Active load trajectories with 5 induction motors.

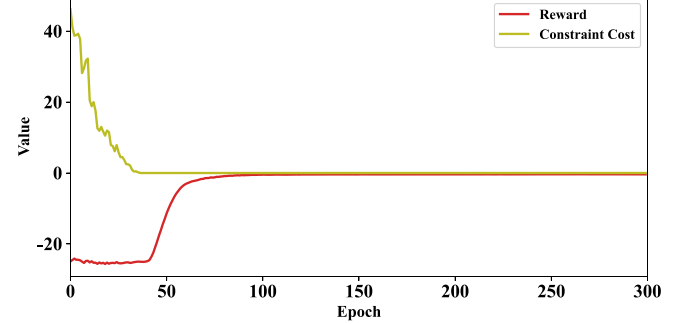


Fig. 17. Reward and constraint cost trajectories of RCPO during the training process considering dynamic loads and LVRT.

voltage duration exceeds its set threshold or drops to an unrecoverable level. This prevents equipment damage and mitigates further system instability. According to IEEE Std. 1547 [46], we configured LVRT relay parameters for both synchronous generators and PV systems. For a random scenario, a fault and TSEC were applied, with the generator status and voltage responses shown in Figs. 14 and 15, respectively.

Dynamic loads in power system transients modify power demand characteristics, influencing transient stability and emergency control. Unlike static loads, dynamic loads—such as induction motors—introduce inertia and time-delay effects under voltage disturbances, which can lead to voltage oscillations, power imbalances, or even voltage collapse. Their response directly impacts the effectiveness of load-shedding and recovery strategies. Therefore, we replaced 5 loads in the microgrid with induction motors at buses 4, 15, 24, 38, and 45, corresponding to L2, L6, L12, L18, and L23. For the same random scenario, the active load trajectories after the fault, with TSEC, are shown in Fig. 16. As observed, following the short circuit and voltage drop, the electromagnetic torque of the induction motors rapidly decreases, causing a sudden reduction in active power, even turning negative in some cases. Compared to static loads, all 5 dynamic loads exhibit more significant power oscillations.

Based on the new configuration, we regenerated the samples and retrained the DSPP model, achieving an A_P of 99.25% and an MAE of 3.25. This represents a slight performance decline compared to previous results, primarily due to the introduction of dynamic loads. However, the accuracy remains within acceptable limits. Subsequently, we retrained the RCPO model, with its reward and constraint cost converging to -0.388 and 0, respectively, as shown in Fig. 17.

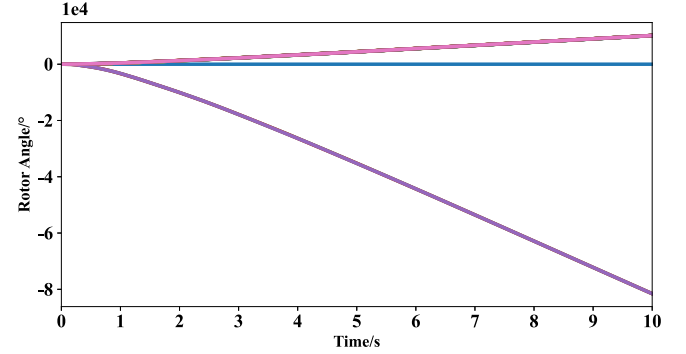


Fig. 18. Rotor angles of all DGs without the TSEC considering dynamic loads and LVRT.

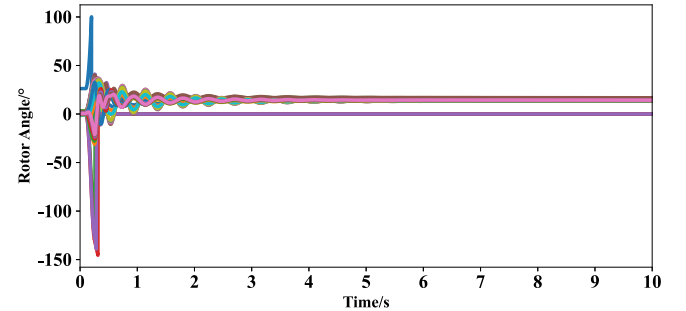


Fig. 19. Rotor angles of all DGs with the TSEC considering dynamic loads and LVRT.

For the same scenario, the TSEC is performed, and the TSI changes from -99.21 to 91.11, as shown in Fig. 18 and 19, respectively, demonstrating that the proposed TSEC method is also applicable to dynamic loads and LVRT.

V. CONCLUSION

This paper proposes a safe RL-based chance-constrained TSEC method for islanded microgrids considering topology reconfigurations. The output of DSPP is the normal distribution of TSE, which is then transformed into a chance constraint, significantly accelerating the TSE speed. RCPO incorporates penalized constraints into the reward and trains through a three-timescale parameter update to find a TSEC strategy that satisfies constraints. Test results on a real microgrid demonstrate that, compared to other RL, safe RL, and sensitivity-based methods, the proposed method can quickly and accurately obtain the optimal load-shedding strategy while ensuring transient stability by controlling DGs, BESS, loads, and switches. In future research, the grid-forming-controlled inverter model will be utilized to study rotor angle stability as well as other types of stability problems.

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